

Linear algebra: Exercise session 5

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1. Are the following matrices invertible? If so, what are their inverses?

$$A = \begin{bmatrix} -3 & 5 & 1 \\ 1 & 6 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}, \quad C = \begin{bmatrix} 4 & 1 & 4 \\ 0 & 1 & 0 \\ 4 & 9 & 4 \end{bmatrix}, \quad D = \begin{bmatrix} 4 & 1 & 2 \\ 2 & 4 & 1 \\ 8 & -7 & 4 \end{bmatrix}, \quad E = \begin{bmatrix} -1 & 5 & 2 \\ 0 & 6 & -1 \\ 1 & 1 & -3 \end{bmatrix},$$

$$F = \begin{bmatrix} -4 & 1 & -2 \\ 2 & -4 & -1 \\ 1 & -2 & -4 \end{bmatrix}, \quad G = \begin{bmatrix} -3 & 1 & -3 \\ 1 & -1 & 2 \\ 2 & 1 & 0 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & -1 & -8 & 0 \\ 0 & 4 & -1 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 7 \end{bmatrix}, \quad I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 7 \end{bmatrix}.$$

Answer.

Matrix A: This matrix is 2×3 , meaning it is not square, and non-square matrices are not invertible.

Matrix B: $a_{11}a_{22} - a_{12}a_{21} = 24 - 14 = 10 \neq 0$, so B is invertible and

$$B^{-1} = \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}.$$

Matrix C: $c_1 = c_3$, so C is not invertible.

Matrix D: $c_1 = 2c_3$, so D is not invertible.

Matrix E: $r_1 + r_3 = r_2$, so E is not invertible.

Matrix F: $|-4| > |1| + |-2|$ and $|-4| > |-1| + |2|$, so F is strictly diagonally dominant matrix and it is invertible.

$$\begin{aligned} [F|I] &= \left[\begin{array}{ccc|ccc} -4 & 1 & -2 & 1 & 0 & 0 \\ 2 & -4 & -1 & 0 & 1 & 0 \\ 1 & -2 & -4 & 0 & 0 & 1 \end{array} \right] \xrightarrow{r_1 \rightarrow -\frac{1}{4}r_1} \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} & 0 & 0 \\ 2 & -4 & -1 & 0 & 1 & 0 \\ 1 & -2 & -4 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{r_2 \rightarrow r_2 - 2r_1, \\ r_3 \rightarrow r_3 - r_1}} \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} & 0 & 0 \\ 0 & -\frac{7}{2} & -2 & \frac{1}{2} & 1 & 0 \\ 0 & -\frac{7}{4} & -\frac{9}{2} & \frac{1}{4} & 0 & 1 \end{array} \right] \\ &\xrightarrow{r_2 \rightarrow -\frac{2}{7}r_2} \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{4} & \frac{1}{2} & -\frac{1}{4} & 0 & 0 \\ 0 & 1 & \frac{4}{7} & -\frac{1}{7} & -\frac{2}{7} & 0 \\ 0 & -\frac{7}{4} & -\frac{9}{2} & \frac{1}{4} & 0 & 1 \end{array} \right] \xrightarrow{\substack{r_3 \rightarrow r_3 + \frac{7}{4}r_2, \\ r_1 \rightarrow r_1 + \frac{1}{4}r_2}} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{9}{14} & -\frac{2}{7} & -\frac{1}{14} & 0 \\ 0 & 1 & \frac{4}{7} & -\frac{1}{7} & -\frac{2}{7} & 0 \\ 0 & 0 & -\frac{7}{2} & 0 & -\frac{1}{2} & 1 \end{array} \right] \\ &\xrightarrow{r_3 \rightarrow -\frac{2}{7}r_3} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{9}{14} & -\frac{2}{7} & -\frac{1}{14} & 0 \\ 0 & 1 & \frac{4}{7} & -\frac{1}{7} & -\frac{2}{7} & 0 \\ 0 & 0 & 1 & 0 & \frac{1}{7} & -\frac{2}{7} \end{array} \right] \xrightarrow{\substack{r_1 \rightarrow r_1 - \frac{9}{14}r_3, \\ r_2 \rightarrow r_2 - \frac{4}{7}r_3}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{2}{7} & -\frac{8}{49} & \frac{9}{49} \\ 0 & 1 & 0 & -\frac{1}{7} & -\frac{18}{49} & \frac{8}{49} \\ 0 & 0 & 1 & 0 & \frac{1}{7} & -\frac{2}{7} \end{array} \right] \end{aligned}$$

Thus,

$$F^{-1} = \begin{bmatrix} -\frac{2}{7} & -\frac{8}{49} & \frac{9}{49} \\ -\frac{1}{7} & -\frac{18}{49} & \frac{8}{49} \\ 0 & \frac{1}{7} & -\frac{2}{7} \end{bmatrix}.$$

Matrix G: We apply Gauss-Jordan Elimination:

$$\begin{aligned}
[G|I] &= \left[\begin{array}{ccc|ccc} -3 & 1 & -3 & 1 & 0 & 0 \\ 1 & -1 & 2 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{r_1 \rightarrow -\frac{1}{3}r_1} \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{3} & 1 & -\frac{1}{3} & 0 & 0 \\ 1 & -1 & 2 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{r_2 \rightarrow r_2 - r_1, \\ r_3 \rightarrow r_3 - 2r_1}} \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{3} & 1 & -\frac{1}{3} & 0 & 0 \\ 0 & -\frac{2}{3} & 1 & \frac{1}{3} & 1 & 0 \\ 0 & \frac{5}{3} & -2 & \frac{2}{3} & 0 & 1 \end{array} \right] \\
&\xrightarrow{r_2 \rightarrow -\frac{3}{2}r_2} \left[\begin{array}{ccc|ccc} 1 & -\frac{1}{3} & 1 & -\frac{1}{3} & 0 & 0 \\ 0 & 1 & -\frac{3}{2} & -\frac{1}{2} & -\frac{3}{2} & 0 \\ 0 & \frac{5}{3} & -2 & \frac{2}{3} & 0 & 1 \end{array} \right] \xrightarrow{\substack{r_1 \rightarrow r_1 + \frac{1}{3}r_2, \\ r_3 \rightarrow r_3 - \frac{5}{3}r_2}} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & -\frac{3}{2} & -\frac{1}{2} & -\frac{3}{2} & 0 \\ 0 & 0 & \frac{1}{2} & \frac{2}{3} & \frac{5}{2} & 1 \end{array} \right] \\
&\xrightarrow{r_3 \rightarrow 2r_3} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & -\frac{3}{2} & -\frac{1}{2} & -\frac{3}{2} & 0 \\ 0 & 0 & 1 & \frac{4}{3} & 5 & 2 \end{array} \right] \xrightarrow{\substack{r_1 \rightarrow r_1 - \frac{1}{2}r_3, \\ r_2 \rightarrow r_2 + \frac{3}{2}r_3}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{7}{6} & -3 & -1 \\ 0 & 1 & 0 & \frac{3}{2} & 6 & 3 \\ 0 & 0 & 1 & \frac{4}{3} & 5 & 2 \end{array} \right].
\end{aligned}$$

Thus,

$$G^{-1} = \begin{bmatrix} -\frac{7}{6} & -3 & -1 \\ \frac{3}{2} & 6 & 3 \\ \frac{4}{3} & 5 & 2 \end{bmatrix}.$$

Matrix H : $r_3 = 0$, so H is not invertible.

Matrix I : Matrix I is a diagonal matrix with nonzero diagonal entries, so it is invertible.

$$I^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{7} \end{bmatrix}.$$

2. Solve the following systems of linear equations by using the inverse matrix.

$$\begin{aligned}
\text{(a)} \quad & \begin{cases} 3x_1 + 8x_2 = 5, \\ 4x_1 + 11x_2 = 7. \end{cases} & \text{(b)} \quad \begin{cases} 5x_1 + 15x_2 + 56x_3 = 35, \\ -4x_1 - 11x_2 - 41x_3 = -26, \\ -x_1 - 3x_2 - 11x_3 = -7. \end{cases}
\end{aligned}$$

Answer.

Part(a): $AX = b$, where

$$A = \begin{bmatrix} 3 & 8 \\ 4 & 11 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 7 \end{bmatrix}.$$

First, we find inverse of A , and then, compute $X = A^{-1}b$:

$$A^{-1} = \begin{bmatrix} 11 & -8 \\ -4 & 3 \end{bmatrix}, \quad X = A^{-1}b = \begin{bmatrix} 11 & -8 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \Rightarrow \begin{cases} x_1 = -1, \\ x_2 = 1. \end{cases}$$

Part(b): $AX = B$, where

$$A = \begin{bmatrix} 5 & 15 & 56 \\ -4 & -11 & -41 \\ -1 & -3 & -11 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 35 \\ -26 \\ -7 \end{bmatrix}.$$

First, we find inverse of A , and then, compute $X = A^{-1}b$:

$$\begin{aligned}
[A|I] &= \left[\begin{array}{ccc|ccc} 5 & 15 & 56 & 1 & 0 & 0 \\ -4 & -11 & -41 & 0 & 1 & 0 \\ -1 & -3 & -11 & 0 & 0 & 1 \end{array} \right] \xrightarrow{r_1 \rightarrow \frac{1}{5}r_1} \left[\begin{array}{ccc|ccc} 1 & 3 & 11.2 & 0.2 & 0 & 0 \\ -4 & -11 & -41 & 0 & 1 & 0 \\ -1 & -3 & -11 & 0 & 0 & 1 \end{array} \right] \\
&\xrightarrow{\substack{r_2 \rightarrow r_2 + 4r_1, \\ r_3 \rightarrow r_3 + r_1}} \left[\begin{array}{ccc|ccc} 1 & 3 & 11.2 & 0.2 & 0 & 0 \\ 0 & 1 & 3.8 & 0.8 & 1 & 0 \\ 0 & 0 & 0.2 & 0.2 & 0 & 1 \end{array} \right] \xrightarrow{\substack{r_1 \rightarrow r_1 - 3r_2, \\ r_3 \rightarrow 5r_3}} \left[\begin{array}{ccc|ccc} 1 & 0 & -0.2 & -2.2 & -3 & 0 \\ 0 & 1 & 3.8 & 0.8 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 5 \end{array} \right] \\
&\xrightarrow{\substack{r_1 \rightarrow r_1 + 0.2r_3, \\ r_2 \rightarrow r_2 - 7.8r_3}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & -3 & 1 \\ 0 & 1 & 0 & -3 & 1 & -19 \\ 0 & 0 & 1 & 1 & 0 & 5 \end{array} \right].
\end{aligned}$$

Thus,

$$A^{-1} = \begin{bmatrix} -2 & -3 & 1 \\ -3 & 1 & -19 \\ 1 & 0 & 5 \end{bmatrix}, \quad X = A^{-1}b = \begin{bmatrix} -2 & -3 & 1 \\ -3 & 1 & -19 \\ 1 & 0 & 5 \end{bmatrix} \begin{bmatrix} 35 \\ -26 \\ -7 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \implies \begin{cases} x_1 = 1, \\ x_2 = 2, \\ x_3 = 0. \end{cases}$$

3. Prove the following statements:

- (a) The $n \times n$ matrix A is invertible (nonsingular) if and only if $\text{rank}(A) = n$ if and only if $N(A) = \{0\}$ ($\dim(N(A)) = 0$).
- (b) If A is an $n \times n$ invertible matrix such that $A^2 = A$, then it is the identity matrix.
- (c) If A is an $n \times n$ matrix with $A^n = 0$, for some number n , then it is noninvertible.
- (d) If A is an $n \times n$ matrix with $A^n = 0$, for some number n , then the matrix $(I - A)$ is invertible and its inverse is $(I + A + A^2 + \dots + A^{n-1})$.

Answer.

- (a) The $n \times n$ matrix A is invertible (nonsingular) if and only if applying Gauss-Jordan Elimination method, A is transformed to I_n and has n pivots if and only if $\text{rank}(A) = n$ if and only if $\dim(N(A)) = n - \text{rank}(A) = n - n = 0$ and $N(A) = \{0\}$.
- (b) Let A be an $n \times n$ invertible matrix such that $A^2 = A$. Then,

$$A^2 = A \implies A^{-1}A^2 = A^{-1}A \implies A = I_n.$$

- (c) Let A be an $n \times n$ matrix with $A^n = 0$, for some number n . If $A = 0$, then it clearly is not invertible. So, assume the another case, $A \neq 0$. By indirect proof, suppose that it is invertible. Then,

$$A^n = 0 \implies A^{-1}A^n = A^{-1}0 \implies A^{n-1} = 0 \implies \dots \implies A = 0,$$

which makes a contradiction. Therefore, A is not invertible.

- (d) Let A be an $n \times n$ matrix with $A^n = 0$, for some number n . Then,

$$(I - A)(I + A + A^2 + \dots + A^{n-1}) = I + A + A^2 + \dots + A^{n-1} - (A + A^2 + \dots + A^{n-1} + A^n) = I - A^n = I.$$

So, the matrix $(I - A)$ is invertible and its inverse is $(I + A + A^2 + \dots + A^{n-1})$.

4. Let A, B and X be three square invertible matrices of the same size. Find an expression of X in the following cases:

- (a) $5A - X = I - 5X$.
- (b) $2AX^{-1} + B = 0$.
- (c) $AXB = A + B$.
- (d) $AX^{-1} + X^{-1} = B$.

Answer. We try to find X based on A, B :

- (a) $5A - X = I - 5X \implies X = \frac{1}{4}(I - 5A)$.
- (b) $2AX^{-1} + B = 0 \implies 2AX^{-1} = -B \implies X = -2B^{-1}A$.
- (c) $AXB = A + B \implies AX = AB^{-1} + I \implies X = B^{-1} + A^{-1}$.
- (d) $AX^{-1} + X^{-1} = B \implies A + I = BX \implies X = B^{-1}(A + I)$.

5. Consider the following matrix M , where A is a $q \times p$ and B is an $r \times q$ matrix. What is the inverse of M and N ?

$$M = \begin{bmatrix} I_p & 0 \\ A & I_q \end{bmatrix}, \quad N = \begin{bmatrix} I_p & 0 & 0 \\ A & I_q & 0 \\ 0 & B & I_r \end{bmatrix}.$$

Answer.

Matrix M :

$$[M|I] = \left[\begin{array}{cc|cc} I_p & 0 & I_p & 0 \\ A & I_q & 0 & I_q \end{array} \right] \xrightarrow{r_2 \rightarrow r_2 - Ar_1} \left[\begin{array}{cc|cc} I_p & 0 & I_p & 0 \\ 0 & I_q & -A & I_q \end{array} \right].$$

Thus,

$$M^{-1} = \begin{bmatrix} I_p & 0 \\ -A & I_q \end{bmatrix}.$$

Matrix N :

$$\begin{aligned} [N|I] &= \left[\begin{array}{ccc|ccc} I_p & 0 & 0 & I_p & 0 & 0 \\ A & I_q & 0 & 0 & I_q & 0 \\ 0 & B & I_r & 0 & 0 & I_r \end{array} \right] \xrightarrow{r_2 \rightarrow r_2 - Ar_1} \left[\begin{array}{ccc|ccc} I_p & 0 & 0 & I_p & 0 & 0 \\ 0 & I_q & 0 & -A & I_q & 0 \\ 0 & B & I_r & 0 & 0 & I_r \end{array} \right] \\ &\xrightarrow{r_3 \rightarrow r_3 - Br_2} \left[\begin{array}{ccc|ccc} I_p & 0 & 0 & I_p & 0 & 0 \\ 0 & I_q & 0 & -A & I_q & 0 \\ 0 & B & I_r & BA & -B & I_r \end{array} \right]. \end{aligned}$$

Thus,

$$N^{-1} = \begin{bmatrix} I_p & 0 & 0 \\ -A & I_q & 0 \\ BA & -B & I_r \end{bmatrix}.$$

6. Compute the inverse for the following matrices

$$L = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ c & b & 1 \end{bmatrix}, \quad D = \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & a' & c' \\ 0 & 1 & b' \\ 0 & 0 & 1 \end{bmatrix}$$

where $d_1, d_2, d_3 \neq 0$.

Is there an easy way to compute the inverse of any matrix A which can be written as $A = LDU$?

Answer.

Matrix L :

$$\begin{aligned} [L|I] &= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ a & 1 & 0 & 0 & 1 & 0 \\ c & b & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{r_2 \rightarrow r_2 - ar_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -a & 1 & 0 \\ c & b & 1 & 0 & 0 & 1 \end{array} \right] \\ &\xrightarrow{r_3 \rightarrow r_3 - cr_1 - br_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -a & 1 & 0 \\ 0 & 0 & 1 & -c + ab & -b & 1 \end{array} \right]. \end{aligned}$$

Thus,

$$L^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -a & 1 & 0 \\ ab - c & -b & 1 \end{bmatrix}.$$

Matrix D :

$$D = \begin{bmatrix} \frac{1}{d_1} & 0 & 0 \\ 0 & \frac{1}{d_2} & 0 \\ 0 & 0 & \frac{1}{d_3} \end{bmatrix}.$$

Matrix U :

$$\begin{aligned} [U|I] &= \left[\begin{array}{ccc|ccc} 1 & a' & c' & 1 & 0 & 0 \\ 0 & 1 & b' & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{r_1 \rightarrow r_1 - c'r_3, \\ r_2 \rightarrow r_2 - b'r_3}} \left[\begin{array}{ccc|ccc} 1 & a' & 0 & 1 & 0 & -c' \\ 0 & 1 & 0 & 0 & 1 & -b' \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \\ &\xrightarrow{r_1 \rightarrow r_1 - a'r_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -a' & -c' + a'b' \\ 0 & 1 & 0 & 0 & 1 & -b' \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]. \end{aligned}$$

Thus,

$$U^{-1} = \begin{bmatrix} 1 & -a' & a'b' - c' \\ 0 & 1 & -b' \\ 0 & 0 & 1 \end{bmatrix}.$$

The easy way is to compute $A^{-1} = U^{-1}D^{-1}L^{-1}$.